

EXECUTIVE INVESTMENT MEMORANDUM

PROJECT TITAN: 2-GIGAWATT AI DATA CENTER (TEXAS ERCOT REGION)

Model Architect: *DeepSeek R1 Distill Qwen 14B* | Benchmark Run: *October 2024* | Classification: *Tier-IV Mission Critical*

This document provides an institutional investment analysis evaluating the capital deployment, civil engineering feasibility, power interconnection, and financial return profile for constructing a **2.0 Gigawatt AI hyperscale facility** in Texas. Based on the economic projections synthesized by **DeepSeek R1 Distill Qwen 14B**, total Capital Expenditure (CapEx) is modeled at **\$962,500,000** (Base Range: \$925,000,000 to \$1,000,000,000), with annualized Operational Expenditures (OpEx) of **\$100,000,000/year**. The facility is engineered to support ultra-dense AI clusters exceeding 100 kW/rack with redundant grid feeds.

Total Planned Capacity	Target Facility Area	Estimated CapEx (Mid)	Est. Annual OpEx	Target PUE
2.0 GW (2,000 MW)	1,000,000 sq ft	\$962,500,000	\$100,000,000	1.15 - 1.25

Capital Expenditure Allocation & Cost Centers

Item	Cost Center Description	Amount (\$USD)	Share (%)
1	Dallas-Fort Worth Land Acquisition	\$15,000,000	1.1%
2	Tier IV Building Construction (\$350/sq ft)	\$350,000,000	25.5%
3	Power Distribution & Chiller Systems	\$150,000,000	10.9%
4	High-Performance Servers & Networking	\$250,000,000	18.2%
5	Permits, Environmental & Training Programs	\$40,000,000	2.9%
6	Contingency Reserve (10%)	\$70,000,000	5.1%
7	5-Year Cumulative Operating Reserve	\$500,000,000	36.4%
TOTAL CAPITAL INVESTMENT		\$1,375,000,000	100.0%

Strategic Architecture & Critical ERCOT Grid Risk Analysis

- 1. ERCOT Substation Interconnection:** Interconnecting a continuous 2.0 GW load requires direct 345 kV dual-circuit transmission lines, on-site utility substations, and multi-year coordination with ERCOT grid operators. Peak summer heat and grid curtailment necessitate on-site generation and battery backup.
- 2. Advanced Cooling Architecture:** With high-density AI accelerators exceeding 1 kW per socket, standard air cooling is insufficient. The model specifies closed-loop liquid-to-liquid heat exchangers and immersion manifolds, achieving an optimized PUE while conserving scarce regional water reserves.
- 3. Investment Recommendation:** The economics confirm a highly defensible infrastructure asset. Initial phasing in 250 MW to 500 MW data hall increments provides de-risked capital deployment and allows forward cash flows to fund sequential phases.